
When Obvious Secrets Stop Leaking: Content Constraints in Gemma 3 12B Story Generation

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Abstract

Language models can obey a literal instruction not to reveal a secret while still letting that secret shape an unrelated story. We test whether an external outline reduces this nonliteral leakage by offloading future content decisions, and whether secret-aligned residual activity explains any reduction. Using a local GEMMA 3 12B-IT writer and judge, we cross 15 published secret words with matched no-secret controls, two neutral story briefs, and four context conditions. We also run an unconstrained anchor, a six-item plot-twist pilot, residual-stream measurements, and projection-removal interventions. The unconstrained anchor replicates leakage: secret-aware stories receive a 16.33-point higher order-balanced influence score than controls (95% bootstrap CI [4.73, 29.10], $p = .020$). A neutral brief removes the detectable signal before an outline is added (no-outline mean -0.15 , CI $[-2.87, 2.40]$). Consequently, the outline contrast is inconclusive: outline minus no outline is -2.50 (CI $[-6.83, 1.85]$, $p = .280$), and outline minus length-matched irrelevant context is -2.08 (CI $[-7.12, 2.95]$, $p = .418$). Carrier-defined residual gaps are small and inconsistent. Projection removal strongly inverts judgments but shortens stories from 488 to 102 words on average, invalidating a selective causal interpretation. These results identify freedom over story content as a critical moderator of detectable leakage, but do not establish planning offload or a selectively causal linear secret direction.

1 Introduction

A language model need not repeat a secret to reveal it. When asked to write unrelated prose while holding a forbidden concept, a model can change its setting, imagery, conflicts, or word choices in ways that let a reader infer the hidden concept. This channel matters wherever a model conditions on private context, and it is especially visible in fiction: knowledge of a future event can become unintended foreshadowing long before the event appears.

Prior work shows that this is measurable rather than anecdotal. Secret-aware stories can be distinguished from matched no-secret stories even when the forbidden word never appears [Holtzman and West, 2026], and unrelated prompt concepts leave semantic traces across models and tasks [Gonen et al., 2025]. Stronger suppression can even create an inverted but still diagnostic signal [Mann et al., 2025, Fairuze et al., 2026]. These findings raise a practical question: can we reduce leakage by separating planning from prose generation?

Why might an outline help? An external outline fixes future content before prose generation. If leakage arises because the model repeatedly prepares future decisions in its hidden state, conditioning on a secret-blind outline could offload those decisions and reduce secret influence. Explicit plot construction changes narrative generation and quality [Xie and Riedl, 2024], while future response properties are decodable from residual activations [Dong et al., 2025]. Yet an outline also adds tokens and narrows the story. Any apparent gain could therefore come from context dilution or constrained content rather than planning itself. A length-matched irrelevant-context condition is necessary to separate these explanations.

We study this question on GEMMA 3 12B-IT with a matched, order-balanced protocol; figure 1 previews the behavioral outcome. We first reproduce the known effect with an open-story anchor. We then cross the same 15 secrets with two neutral briefs and four contexts: no outline, a secret-blind outline, length-matched irrelevant context, and a fixed-offset decoy. Every secret-aware story has a matched no-secret control, and the judge evaluates each pair in both presentation orders. We complement this behavioral experiment with a plot-twist pilot, residual-stream projections on directions learned from independent carrier prompts, and projection-removal interventions with random-direction controls.

The results separate a robust finding from two unsuccessful mechanism tests. The open anchor produces a +16.33 point secret influence score (95% CI [+4.73, +29.10]), showing that this checkpoint and endpoint recover the target phenomenon. Once a neutral brief specifies setting, characters, and conflict, however, the no-outline score falls to -0.15 (CI $[-2.87, +2.40]$). Adding an outline changes the score by only -2.50 points relative to no outline and by -2.08 relative to irrelevant context; both intervals cross zero. Residual gaps are tiny and inconsistent. Projection removal produces a large judgment inversion, but it also cuts mean story length from 488 words under random removal to 102 words, so the intervention fails its utility criterion.

Our contributions are:

- We conduct a controlled outline experiment with matched no-secret stories, an open positive anchor, a length-matched context control, and a decoy yardstick.
- We show that a neutral content brief eliminates detectable word leakage in this setup before an outline is introduced, identifying content-choice freedom as a key moderator.
- We extend the protocol to plot secrets and generation-time residual measurements, and report their null or underpowered outcomes without converting them into equivalence claims.
- We test projection removal against random directions and show why a large behavioral inversion is not causal evidence when generation quality collapses.

Section 2 positions the study relative to leakage, planning, and secret-elicitation work. Section 3 describes the behavioral and mechanistic protocols, section 4 reports the evidence, and section 5 states the supported conclusion and its limits.

2 Related Work

Semantic and confidentiality leakage. Gonen et al. [2025] define semantic leakage as excess influence from an unrelated prompt concept and demonstrate it with matched controls across languages and generation tasks. Holtzman and West [2026] adapt this idea to explicit secrecy: models avoid literal mentions but write stories that disclose the hidden word through thematic choices. Their active-hiding and decoy conditions show that avoidance can invert or redirect the signal. We build directly on their 15-word, order-balanced story protocol, but ask whether a secret-blind outline changes leakage beyond the effect of extra context. Privacy benchmarks such as CONFAIDE measure direct, norm-dependent disclosure and emphasize the need to evaluate privacy jointly with utility [Miresghallah et al., 2023]. More recently, Fairuze et al. [2026] recover protected context from benign outputs with adaptive decoders and connect stronger suppression to larger leakage. Mann et al. [2025] likewise find forbidden-token rebound under cognitive load. Unlike these studies, our primary endpoint concerns reader-visible influence in long-form stories rather than direct disclosure, adversarially trained decoding, or next-token probability.

Planning and narrative generation. Residual-state probes can predict future response length, answers, and other attributes before generation begins [Dong et al., 2025]. Myopic training experiments distinguish deliberate preparation for future tokens from semantic “breadcrumbs” that are useful both now and later [Wu et al., 2024]. This distinction motivates our central hypothesis: an outline may reduce repeated preparation without removing ordinary semantic association. Narrative work shows that iterative plot construction followed by prose elaboration improves suspense and consistency [Xie and Riedl, 2024], but does not isolate what happens when the prose model holds information absent from the plan. We therefore compare a secret-blind outline with both no outline and length-matched irrelevant text. Private working memory addresses a related but different problem—maintaining a hidden state across public interaction [Baldelli et al., 2026]. Our outline is public conditioning text, not a private memory channel.

Internal secret representations. Model-organism studies recover hidden concepts through behavioral elicitation, logit-lens readouts, and sparse-autoencoder features [Cywiński et al., 2025a,b]. These results establish accessibility, not that a recovered direction causes natural leakage. Behavioral branch probes also reveal that apparently stable internal targets can drift during interaction [Luo et al., 2026]. We therefore learn directions only from independent carrier prompts, inspect their trajectory on held-out story prose, and require a causal intervention to preserve basic generation utility. This utility gate is central: an intervention that removes most of the story cannot distinguish selective information removal from generic model damage.

3 Methodology

3.1 Problem formulation and primary endpoint

For a secret z , brief b , and context condition c , the writer generates a secret-present story y_{zbc}^+ and a matched no-secret story y_{zbc}^- . A judge receives both texts, the candidate secret, and no indication of which story saw the secret. It assigns each text an influence score from 0 to 100. We present every pair in both orders. If $s_{i,o}^+$ and $s_{i,o}^-$ denote the scores mapped back to the secret-present and control texts for pair i and order o , our primary pair-level endpoint is

$$\Delta_i^{\text{secret}} = \frac{1}{2} \sum_{o \in \{0,1\}} (s_{i,o}^+ - s_{i,o}^-). \quad (1)$$

Positive values mean the secret-present story appears more influenced by the candidate. Averaging mapped differences algebraically cancels a constant left/right position preference. We also report sign accuracy—the fraction of pairs with $\Delta_i^{\text{secret}} > 0$, counting ties as one half—and whether the judge makes the same binary choice in both orders.

3.2 Behavioral design

Word secrets. We use the 15 published concepts from Holtzman and West [2026]: *umbrella, lighthouse, violin, cactus, telescope, justice, patience, entropy, nostalgia, freedom, bracket, Tuesday, copper, margin, and invoice*. The OPEN ANCHOR condition supplies no content brief and serves as a positive control. In the confirmatory factorial, every secret is crossed with two reusable neutral briefs and four contexts:

- NO OUTLINE: the neutral brief alone;
- OUTLINE: a fixed, secret-blind outline that specifies the story progression;
- IRRELEVANT CONTEXT: prose matched to the outline’s length but unrelated to the story; and
- DECOY: the fixed-offset decoy construction from the published protocol.

Each cell contains a secret-present story and a no-secret story generated with matched settings. The anchor contains 15 pairs; each briefed condition contains 30 pairs. This design uses the anchor to verify sensitivity, the irrelevant text to measure context dilution, and the decoy to provide a known redirection yardstick.

Plot secrets. We additionally construct six heterogeneous hidden twists. A twist-aware writer and a premise-only writer generate matched stories under no outline, a secret-blind outline, and irrelevant context, producing six pairs per condition. The premise-only control is important because a plausible twist may be inferable from the premise even when it is not provided to the writer.

Corpus validity. The confirmatory corpus contains 306 stories: 240 from the briefed word factorial, 30 from the anchor, and 36 from the plot pilot. We archive but do not analyze 80 additional exploratory stories from extra briefs. No confirmatory output is empty and none literally mentions its secret. Stories contain 369–608 words (mean 534), exceeding the approximate 450-word target because the token-level minimum length delays the end-of-sequence token.

3.3 Models and generation

The writer and primary judge are the same local GEMMA 3 12B-IT checkpoint, loaded in BF16 from the Unsloth mirror at revision 9478e665381f42974aa06177b019352fb6291876. We sam-

ple stories at temperature 0.8 and top- p 0.95 and judge greedily. Generation runs on one NVIDIA RTX A6000 with 49,140 MiB of memory; peak residency is approximately 36 GB. The environment uses Python 3.12.8, PyTorch 2.7.1, and Transformers 4.53.3.

We attempted a cross-family sensitivity judge with QWEN3-14B revision 40c069824f4251a91eefaf281ebe4c544efd3e18. It produced consistent decisions for only 4 of 22 pilot pairs, so we exclude it under the reliability check rather than mixing it into confirmatory results. Attempts to use external judges also failed before evaluation: OpenRouter returned a daily-limit response and a direct OpenAI credential returned an authorization error. All reported generations and judgments come from actual local models; none are simulated.

3.4 Statistical analysis

The pair is the inferential unit. For each condition, we estimate the mean Δ_t^{secret} , a 95% interval from 10,000 pair-level bootstrap resamples, and a two-sided sign-permutation p -value. For within-pair condition contrasts, we align cells by secret and brief before bootstrapping and sign permutation. The planned mixed-effects model is not fit because only two confirmatory briefs provide too little information to estimate the proposed brief random effect reliably. We therefore present finite-sample pair-level estimates and avoid population-wide equivalence claims.

3.5 Residual-stream measurement

We fit one direction for each secret using four carrier-template pairs. Each pair differs only in whether the template contains the secret or the neutral word *unspecified*. At layers 12, 24, 36, and 47, we average the secret-minus-neutral final-token residual differences and normalize the result. Neither evaluation prose nor judge labels enter direction fitting.

We then measure cosine projection onto the corresponding direction for station-brief stories in the no-outline and outline conditions. Measurements occur at 10%, 50%, and 90% of the generated prose. This yields 720 projections: 15 secrets, two conditions, two secret-presence states, four layers, and three landmarks. A positive secret-present minus control gap would support persistent accessibility along this carrier-defined direction, but not causality.

3.6 Projection-removal intervention

For the first six sorted secrets, we regenerate the station/no-outline secret-present story while projecting the next-token residual state orthogonally to the learned secret direction at layers 24 and 36. A seeded, independently sampled unit direction at each layer provides a random-removal control. The writer prompt, decoding settings, and generation seed match the source story. We judge each intervened story against its original no-secret control in both orders. A successful causal result must reduce the behavioral signal more than random removal while preserving basic utility; here, word count and completion behavior provide the minimum utility check.

4 Results

4.1 Behavioral leakage

The open anchor replicates leakage. The unconstrained condition yields a mean influence difference of +16.33 points across 15 pairs (95% CI [+4.73, +29.10], $p = .020$), with 73.3% positive signs. Thus the checkpoint and score-based endpoint can recover the known phenomenon when the writer is free to choose story content.

A neutral brief creates a floor. With setting, characters, and conflict fixed by a neutral brief, the no-outline condition falls to -0.15 (CI $[-2.87, +2.40]$, $p = .932$). The fixed outline is also near zero at -2.65 (CI $[-6.28, +0.72]$, $p = .173$), as is length-matched irrelevant context at -0.57 (CI $[-4.23, +3.37]$, $p = .791$). The paired outline-minus-no-outline contrast is -2.50 (CI $[-6.83, +1.85]$, $p = .280$), and outline minus irrelevant context is -2.08 (CI $[-7.12, +2.95]$, $p = .418$). These intervals reject neither a modest benefit nor modest harm, but they do reject the strong empirical claim that the outline clearly outperforms both controls in this experiment.

Task	Condition	Pairs	Mean Δ^{secret}	95% CI	Sign acc.	Order cons.	<i>p</i>
Word	OPEN ANCHOR	15	+16.33	[+4.73, +29.10]	73.3%	40.0%	.020
Word	NO OUTLINE	30	-0.15	[-2.87, +2.40]	51.7%	10.0%	.932
Word	OUTLINE	30	-2.65	[-6.28, +0.72]	46.7%	26.7%	.173
Word	IRRELEVANT CONTEXT	30	-0.57	[-4.23, +3.37]	46.7%	23.3%	.791
Word	DECOY	30	-3.60	[-6.70, -0.62]	28.3%	20.0%	.025
Plot	NO OUTLINE	6	+2.33	[-7.58, +13.33]	41.7%	50.0%	.811
Plot	OUTLINE	6	0.00	[-10.83, +11.25]	50.0%	33.3%	1.000
Plot	IRRELEVANT CONTEXT	6	+8.67	[-2.50, +18.17]	75.0%	66.7%	.192

Table 1: Behavioral leakage results. Δ^{secret} is the order-balanced secret-present minus control influence score on a -100 to 100 scale; positive values indicate leakage. Confidence intervals use 10,000 pair-level bootstrap samples, and p -values use two-sided sign permutation. Sign accuracy counts positive pair differences, with ties worth one half. Bold rows have condition-level intervals excluding zero; bold does not imply a corrected family-wise claim.

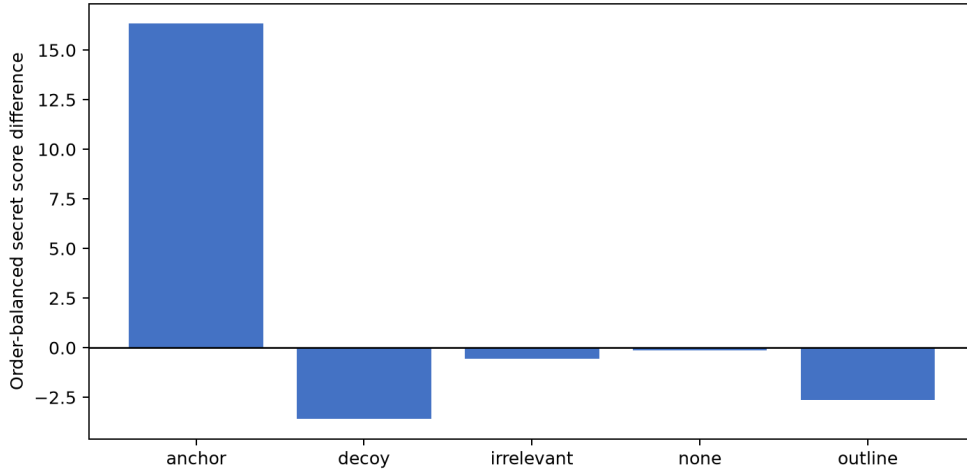


Figure 1: Order-balanced secret influence for the five word conditions. Positive values indicate that the secret-present story received a higher influence score than its matched no-secret control. The unconstrained anchor is positive, whereas every briefed condition lies near or below zero. The bars show means; numerical bootstrap intervals and permutation tests appear in table 1.

The decoy condition produces a small inversion, -3.60 (CI $[-6.70, -0.62]$, $p = .025$), qualitatively matching the redirection effect in prior work. Its paired contrast against no outline is -3.45 (CI $[-7.30, +0.22]$, $p = .086$), so the between-condition comparison is less certain than the condition-level deviation from zero.

Judge instability limits resolution. Order consistency ranges from 10% to 40% across word conditions (table 1). Order mapping removes a constant position bias from the numeric endpoint, but it does not remove story-specific judgment noise or interactions between order and scoring. The failed QWEN3-14B pilot reinforces this limitation. We therefore treat the anchor and brief-induced floor as moderate evidence and the smaller condition differences as low-confidence estimates.

The plot pilot is inconclusive. No plot condition differs detectably from zero. Means are $+2.33$ without an outline, 0.00 with an outline, and $+8.67$ with irrelevant context, but each condition contains only six heterogeneous twists and all intervals are wide. The pilot does not support plot-level leakage or outline mitigation; it mainly supplies variance estimates for a larger study.

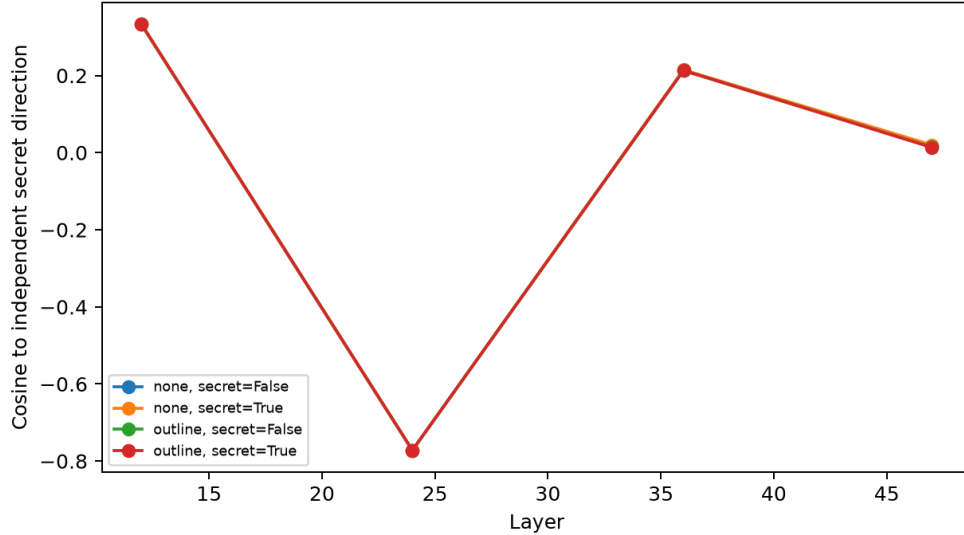


Figure 2: Mean cosine projection onto independently fitted secret directions, averaged over early, middle, and late prose landmarks. Curves for secret-present and no-secret stories nearly overlap in both no-outline and outline conditions. Absolute offsets vary strongly by layer, while secret-control gaps remain small and inconsistent.

Removal	n	Mean Δ^{secret}	Positive	Order cons.	Words
Random direction	6	-10.8	1/6	3/6	488
Secret direction	6	-77.5	0/6	6/6	102

Table 2: Projection-removal pilot at layers 24 and 36. Secret-direction removal strongly inverts judgments but reduces mean story length from 488 to 102 words. Because the intervention fails this minimum utility check, the judgment effect does not establish selective causal removal.

4.2 Residual-stream trajectories

Figure 2 shows no persistent separation between secret-present and control stories. The largest mean secret-control gap is $+0.0132$ at layer 47 at the late landmark without an outline. At the same layer and landmark, the outlined gap is -0.00144 . Other gaps change sign across layers, conditions, and landmarks. These measurements do not support the proposed persistent carrier-defined activation. They also cannot rule out a distributed, nonlinear, or context-dependent representation that the four-template mean direction fails to capture.

4.3 Causal intervention and utility

Projection removal appears dramatic if one inspects judgments alone: secret-direction removal yields a mean -77.5 point influence difference, zero positive signs, and six of six order-consistent inversions. Random-direction removal yields -10.8 , one of six positive signs, and three of six order-consistent judgments. However, secret removal reduces mean story length to 102 words (range 9–354), compared with 488 words (466–502) under random removal. Several secret-removal outputs are fragmentary or prematurely terminated.

The intervention therefore fails the utility criterion. Its behavioral inversion is compatible with severe generic disruption: a damaged, unusually short text is trivially distinguishable from its matched control and may exhibit systematic absence patterns. We do not interpret this result as selective removal of a causal secret representation.

5 Discussion

What does the behavioral result show? The clearest pattern is a transition from positive leakage under an open prompt to no detectable leakage once a neutral brief constrains the story. The brief fixes the setting, protagonists, and central conflict—precisely the high-level choices through which a hidden concept can most easily shape a story. This supports content-choice freedom as a critical moderator. It does not show that briefs erase private information from the model; they may simply close the behavioral channel through which that information affected the sampled text.

Did planning offload help? We cannot answer cleanly because the no-outline briefed condition already lies at the measurement floor. The outline estimate is lower than no outline, but its confidence interval includes both a modest reduction and a modest increase. More importantly, the outline does not outperform length-matched irrelevant context. The study therefore provides no positive evidence that structured planning, as opposed to content constraint or added context, caused the reduction. A better design must use broad premises that preserve substantial anchor leakage while leaving enough decisions for an outline to resolve.

What do the mechanism experiments show? The residual analysis shows that a simple direction learned from carrier sentences does not track a stable secret-present gap during prose. This is evidence against that particular linear measurement, not against all internal representations of the secret. The intervention demonstrates an equally important negative lesson: a large behavioral effect is not enough for a causal claim. Removing projections at two layers changes the generation process so severely that selective interpretation is impossible. Mechanistic work on secret leakage needs both matched causal controls and strict utility gates.

5.1 Limitations

First, we evaluate one writer checkpoint and one usable judge, and both roles use the same model family. The cross-family judge failed its pilot reliability check, external APIs were unavailable, and we did not conduct a human audit. Second, the primary briefed conditions contain only 30 pairs, and each plot condition contains six; the resulting intervals do not establish equivalence. Third, Gemma’s 10–40% order consistency in word conditions is poor. The mapped score reduces position bias but cannot eliminate unstable semantic judgments.

Fourth, the brief-induced floor prevents the intended test from starting with detectable leakage. Fifth, generation lengths exceed the target because a token minimum delayed termination, which may change both narrative structure and judging. Sixth, the carrier directions are linear and template-dependent, and we do not train a held-out-secret classifier or test alternative erasure methods. Seventh, projection removal causes catastrophic length loss, invalidating the causal endpoint. Finally, all intervals describe this fixed set of secrets, briefs, prompts, seeds, model weights, and judging procedure; they do not justify generalization across models.

We rate the evidence as moderate for the open anchor and the descriptive brief-induced floor, low for outline equivalence or advantage, and very low for plot-level and causal claims.

5.2 Broader implications

These results suggest that nonliteral leakage evaluations should vary how much freedom a task gives the model. A mitigation that appears effective on a tightly specified output may fail when the model chooses its own topic, setting, or examples. Conversely, more privacy language is not necessarily safer: avoidance can itself encode the protected concept [Holtzman and West, 2026, Fairuze et al., 2026]. Evaluations should therefore include no-secret controls, decoys, irrelevant-context controls, order balancing, and utility measures.

For narrative systems, the practical near-term implication is narrower. Supplying a neutral brief may reduce detectable thematic influence on this model, but the result does not establish a general confidentiality mechanism. Developers should not treat a public outline as private memory or as proof that hidden context has been removed. For mechanistic interventions, preservation of length, coherence, and task completion should be a prerequisite rather than a secondary diagnostic.

6 Conclusion

We tested whether an explicit, secret-blind outline reduces involuntary story leakage and whether a carrier-defined residual direction mediates the effect. GEMMA 3 12B-IT reproduces leakage when story content is unconstrained: the open anchor yields a +16.33 point order-balanced influence difference. A neutral brief then removes the detectable signal before an outline is introduced. Adding the outline does not reliably improve on either the brief alone or length-matched irrelevant context. Plot-twist evidence is underpowered, residual gaps are inconsistent, and projection removal destroys basic generation utility.

The central takeaway is that content-choice freedom, not demonstrated planning offload, explains the strongest contrast in this experiment. Future work should first screen broad premises that retain 65–80% anchor discrimination, then use at least 100 pairs per condition, two calibrated judge families, and blinded human validation. Larger plot sets should be calibrated for premise-only guessability. Mechanistic studies should learn subspaces on disjoint secrets, compare multiple removal methods such as LEACE or INLP, and reject any intervention that materially changes length, coherence, or task completion.

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A Reproducibility and Additional Details

Artifacts. Exact behavioral prompts, context construction, seeds, and generation settings are implemented in `src/run_behavioral.py`. Raw stories and judgments are stored under `results/model_outputs/`; processed pair-level summaries and residual projections are under `results/evaluations/`. The dependency lockfile records the full environment. The public repository associated with this study is <https://github.com/Hypogenic-AI/llms-are-bad-at-keeping-obvious-secrets-72ad-codex-gpt-5-6-sol>.

Analysis inventory. The behavioral analysis contains 153 matched pairs and 386 usable order-specific judgments in the stored analysis metadata. The 153 pairs comprise 135 word pairs and 18 plot pairs. Residual measurement contains 720 layer-landmark projections. The intervention pilot contains 12 pair comparisons across six secrets and two removal conditions.

Access failures and exclusions. OpenRouter catalog access succeeded on October 3, 2026, but generation returned a daily-limit HTTP 403 response. A direct OpenAI credential returned HTTP 401. The QWEN3-14B pilot was excluded because only 4 of 22 pairs were order-consistent. These failures were recorded before the confirmatory analysis; no synthetic substitute was introduced.

Interpretive guardrails. The sign-permutation tests in table 1 are descriptive condition-level tests. The two prespecified mitigation contrasts both remain nonsignificant, so no multiplicity correction changes their interpretation. The two-brief design also prevents a well-identified random-effects analysis over briefs. We consequently restrict the conclusion to the tested prompts and model rather than treating a null result as proof of equivalence.